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Making Every Meeting Count: A Qualitative Investigation of Multiteam Meeting Events and Their Role in Supporting Coordinated Cancer Care Delivery

Annamaria V. Wolf, MS[1]; Katelyn N. Hedrick, MA[1]; Sydney R. Begerowski, BA[1]; Donald W. Wiper III, MD[2]; Dorothy R. Carter, PhD[3]; and Marissa L. Shuffler, PhD[1]

QUESTION ASKED: How can inpatient cancer care providers use patient-focused meetings, such as rounds and huddles, as opportunities to facilitate cross-team coordination across the complex multiteam system (MTS) that is responsible for patient care?

SUMMARY ANSWER: During meeting discussions, providers can directly consider cross-team coordination by *informing* about activities in different teams, *deliberating* about how different teams are involved in patient care matters, *strategizing* such that future cross-team interactions are incorporated into plans, and leveraging *pedagogy* as an opportunity to create shared understanding across multiple teams. In addition, patient-focused meetings can be designed to include representatives from multiple teams within the MTS (eg, the nursing team and the physician team), and/or team representatives can attend multiple meetings to create communication channels between teams.

WHAT WE DID: Although interdisciplinary, team-based rounds and huddles are increasingly leveraged within cancer care settings, these meetings often neglect to consider that cancer care depends on coordination across a complex constellation of *multiple*, specialized yet interdependent component teams assembled into a MTS. We conducted a single-site, case study involving interviews and observations to better understand cancer care meetings through a MTS lens. The research context was an inpatient oncology unit operating inside a teaching hospital in the Southern United States. Researchers conducted interviews with five providers of gynecologic oncology to gain insight into the MTS structure, which included a physician team, a nursing team, and a case management team. Subsequently, researchers conducted 30 hours of observations of two types of meetings—rounds and huddles. Observations revealed the composition, focus, and structure of the meetings and how the meetings acknowledged or addressed cross-team coordination. Researchers qualitatively analyzed transcripts of observational field notes using a three-phase coding scheme, which resulted in a thematic framework of MTS meeting processes and related subprocesses.

WHAT WE FOUND: Despite differing in their composition, focus, and structure, rounds and huddle meetings exhibited similar processes and subprocesses, some of which were leveraged to consider MTS issues and facilitate cross-team coordination. In addition to meeting processes, MTS considerations were sometimes addressed through meeting composition, most notably when meeting attendees included representatives from multiple component teams of the MTS.

BIAS, CONFOUNDING FACTORS, DRAWBACKS: The present research used a qualitative approach to investigate a single cancer care MTS in depth. Therefore, the structure and characteristics of the MTS under study, and the meetings embedded within, should not necessarily be generalized to other in-hospital oncology settings.

REAL-LIFE IMPLICATIONS: Cancer care delivery relies on effective coordination across multiple specialized teams that operate as a MTS. This research demonstrates how cancer care meetings can be more effectively used as cross-team cancer care coordination tools in inpatient settings. Along with opportunities, there are clear challenges associated with using such meetings as tools for cross-team coordination. By incorporating knowledge gleaned from broader literature on MTS functioning and meeting science, this article offers practitioners strategies to deal with these challenges and leverage opportunities. Accordingly, incorporating these considerations into clinical practice should help to enhance MTS coordination processes and performance, benefiting providers and receivers of cancer care.

ASSOCIATED CONTENT

Appendix

Author affiliations and disclosures are available with the complete article online.

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SPECIAL SERIES: TEAMS IN CANCER CARE

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Making Every Meeting Count: A Qualitative Investigation of Multiteam Meeting Events and Their Role in Supporting Coordinated Cancer Care Delivery

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abstract

PURPOSE This research considers how cross-disciplinary cancer care meetings can facilitate coordination within the multiteam systems (MTSs) that provide inpatient hospital care. We conducted a series of interviews and observations with members of a single cancer care MTS to address the following research questions: (1) what are the key characteristics of MTS cancer care meetings (with regard to composition, focus, and structure)? and (2) how is cross-team coordination acknowledged and addressed during these meetings?

METHODS In this single-site case study of a MTS operating to provide gynecologic oncology care within a teaching hospital, two types of meetings, called rounds and huddles, were held consistently. We used qualitative methods, including interviews with health care professional subject matter experts and 30 hours of observations of cancer care meetings, and analyzed the data in three stages of qualitative coding.

RESULTS Our analyses resulted in a thematic framework detailing key processes, and subprocesses, identified as central to the activities of observed cancer care meetings. Key processes include information sharing, gaining clarity, strategizing, and pedagogy. Discussions and explanations of this framework showcase the ways in which MTS meetings can bolster cross-team coordination and facilitate MTS activities.

CONCLUSION Inpatient cancer care meetings provide opportunities to facilitate MTS coordination in several ways, yet doing so does not come without challenges. Considering these results together with insights from meeting science and MTS research, this article concludes by putting forward practical recommendations for leveraging opportunities and overcoming challenges to use cancer care meetings as tools to support cross-team coordination.

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INTRODUCTION

Meetings, whether prearranged or spontaneous, occur when at least two individuals gather for structured, work-focused discussions.[1] Inpatient cancer care providers have a longstanding practice of conducting meetings that are focused on establishing and refining care plans for individual patients.[2] For example, rounding in cancer care delivery often involves an interdisciplinary group of providers including an attending physician (AP), resident physicians (RPs), and nurses, who assemble to discuss diagnosis and treatment of patients admitted under an AP's care.[3-5] Hospitals often leverage other daily group meetings, or huddles, to facilitate patient care needs such as discharge and throughput.[6-8] Similar to rounds, huddles can include individuals with diverse specializations such as nursing, management, social work, and medicine, but unlike rounds, huddles tend to focus on a greater number of patients across a unit.[9]

Despite the increased use of group meetings to coordinate cancer care, coordination breakdowns are still frequently reported[10,11] and can lead to staggering financial costs for hospitals and poor patient outcomes.[12,13] In part, coordination failures may persist within systems because of a lack of awareness that cancer care—and the meetings that are intended to facilitate cancer care—depends on effective coordination across a complex *multiteam system* (MTS) composed of multiple distinct component groups or teams, rather than coordination within a single undifferentiated team. In the organizational sciences, MTSs are defined as organizational systems connecting two or more component teams, which although pursuing separate team-level objectives and tasks, are bound by virtue of shared superordinate system-level goals that require cooperation and coordination across all teams.[14] MTS structures offer numerous benefits including multiple

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specializations to solve problems, flexibility, and adaptability to tackle dynamic circumstances and greater resource capacity than single teams.[15,16] However, MTSs also present coordination and collaboration challenges that go beyond those faced by single stand-alone teams. By definition, achieving a shared MTS goal requires *cross-team* coordination and collaboration (ie, interactions between members of different teams). Yet, differences between teams with regard to specialization, training, work processes, norms, goals, or priorities often prevent critical cross-team interactions from occurring.[17]

Importantly, previous research has depicted cancer care delivery in hospitals as epitomizing an MTS structure; multiple disciplines and teams of health care providers must align their efforts to provide high-quality care to shared patients.[18,19] Therefore, enhancing the effectiveness of inhospital group meetings in cancer care (eg, rounds and huddles) may require more direct consideration of *multiteam issues*, such as whether meeting participants represent each of the different component teams or whether plans are focused directly on facilitating cross-team coordination processes. Indeed, within the broader literature on meeting science, which identifies best practices for designing and facilitating meetings,[20] scholars have suggested that meetings can serve as a key mechanism for facilitating cross-team coordination.[21]

The goal of this study is to better understand how MTS concepts and coordination issues are—or could be—incorporated into cancer care meetings. We leveraged a qualitative, single-site case study approach to address two focal research questions:

1. What are the key characteristics of cancer care MTS meetings (ie, who is involved, what is the focus, and how are meetings structured)?
2. How is cross-team coordination acknowledged and addressed during cancer care MTS meetings?

The first question aims to describe the nature of rounding meetings in terms of (1) which individuals participate, (2) which goals and objectives appear to be discussed, and (3) how meetings are structured in terms of time and duration. Cancer care meetings are notoriously different in these respects,[22-24] and these characteristics might differentially affect MTS coordination. The second research question focuses on how meeting interactions and activities specifically address cross-team coordination (ie, the ways in which component teams need to interact to achieve overarching superordinate goals). By addressing these two questions, we aim to provide the clinical community with insight into how cancer care meetings can better serve as a mechanism to facilitate MTS coordination and performance.

METHODS

We leveraged interviews and observations of subject matter experts (SMEs) to apply an abductive lens for understanding how MTS issues are (or could be) incorporated in group meetings during inhospital care. An abductive research approach is suitable for exploring nascent lines of inquiry like the one pursued in this research.[25(p138)] We draw from Charmaz's[25] constructivist approach to qualitative research, which allows research questions and focus to evolve and become refined on the basis of new information, knowledge, and data gained during the research process.

Research Context

The research was conducted in collaboration with clinical leaders working in a 20-bed inpatient unit embedded in a southeastern US teaching hospital. Approval for this human participants research was granted by this hospital's institutional review board [IRB] before study launch. Observations and interviews were only conducted with health care providers on the unit who provided informed consent to participate in our study. Two types of meetings occurred almost daily in this context: (1) *rounds* led by APs for patients with gynecologic cancer and (2) *huddles* led primarily by case management professionals within the nursing unit.

Baseline Interviews With SMEs

Before conducting observations, our research team conducted semistructured interviews with five SMEs representing different specialization areas—an AP, nurse navigator (NN), nurse manager, and two registered nurses (RNs). These interviews allowed us to establish a baseline understanding of the research context and identify key areas for meeting observation. Initial questions (Appendix 1, online only) concerned the applicability of the MTS lens to the inpatient gynecologic oncology context and the composition, structure, and purpose of rounding meetings that occurred there. The research team learned about huddles during subsequent observations; huddles were not the focus of the interviews. Interviews lasted approximately 1 hour and were recorded over Zoom. A total of 251 pages of interview transcripts were generated, with individual interviews ranging between 29 and 54 pages, single-spaced. Appendix Figure A1 (online only) illustrates the typical cancer care context as described by SMEs as a complex MTS.

Observations

Approximately 30 hours of observations were made by trained observers over 20 nonconsecutive days during a 6-week period, capturing 16 rounding and 17 huddle episodes. Observers were able to attend rounds and huddles for 13 of 20 days [ie, 65% of the observation time]. Observers took unstructured notes related to observations of attendees, processes, interactions, key events, and meeting start and end times without interfering in the clinicians' work. In total, 95 individual discussions of patient care coordination were observed, with some of the same patients discussed across multiple days and across both rounds and huddles. When possible,

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observations were conducted in pairs to gather multiple perspectives. Two observers were present for 68% of rounds and 41% of huddles. Field notes were transcribed, consolidated, and stored in a secure database accessible only to researchers. Approximately 56 and 10 single-spaced pages of observation transcripts were generated for rounds and huddles, respectively.

Qualitative Analysis

Data were anonymized before analysis. Three coders, who also served as observers, qualitatively analyzed transcripts in three sequential stages of coding as outlined by Charmaz[25] and implemented in similar oncology research contexts.[26,27] The first stage consisted of initial (line-by-line) coding. The second stage used focused coding to identify higher-order themes, which were integrated into a framework that described key cancer care meeting processes (Table 1). In the third stage, coders used the processes and subprocesses from the thematic framework to recode 28 relevant samples from the rounding and huddle transcripts. Intercoder agreement between the three coders was acceptable, *Fleiss' kappa* = 0.731, $P < .0001$. Through additional consensus meetings, coders resolved intercoder discrepancies and made amendments to the thematic framework to reach full agreement on its contents and examples. A detailed description of the qualitative analysis processes is given in Appendix 2 (online only).

RESULTS

Our qualitative analysis focused on addressing our two research questions, which were designed to reveal the ways in which MTS issues are (or could be) incorporated during inhospital group meetings held to facilitate cancer care delivery.

RQ1: What Are the Key Characteristics of Cancer Care MTS Meetings?

To address our first research question, we leveraged the interviews and observations to identify three overarching characteristics of the MTS meetings: (1) *composition* (who is involved?), (2) *focus* (what was the purpose of the meeting?), and (3) *structure* (how was the meeting organized and/or how did the meeting progress over time?). Figure 1 illustrates the typical composition of rounds and huddles.

As illustrated in Figure 1, rounding meetings typically involved the physician team, composed of an AP, a NN, and two to three RPs. RNs were present during approximately 30% of the rounding meetings we observed. It appeared that RNs attended mainly because of happenstance, with APs occasionally making intentional efforts to include them, either in person or via phone. By contrast, huddle meetings typically included both the case management team and the nursing team, the former of which is composed of a case manager and social worker who collaborate to monitor patient admissions, throughput, and discharge progression, whereas the latter includes a nurse manager and several RNs who work to monitor, treat, and care for patients within the unit. Rounds focused on patients with gynecologic cancer admitted to the unit, their care, diagnosis, and treatment. Huddles focused on the same patients in addition to other patients within the unit; topics included patient length of stay and discharge barriers and progress. Rounding start time varied depending on the AP's schedule and typically lasted for 40-90 minutes. Huddles were held at the same time daily, typically lasting for 30 minutes despite discussing more patients than in rounds.

Although rounds and huddles had different compositions, timings, durations, and focus, the results of our qualitative coding also revealed many commonalities in the processes of both MTS meeting types. We summarize these commonalities in our resulting thematic framework of cancer care MTS meeting processes in Table 1. Our framework consists of four high-order process themes—*informing, deliberating, strategizing, and pedagogy*—each of which also included subprocess themes. Broadly, this framework aligns with common and expected cancer care meeting processes; sharing information, making sense of patients' situations, and planning and coordinating care across providers are known components of rounds[3,22] and huddles.[9,23] Furthermore, the first three high-order processes are consistent with general processes known to occur within meetings across other contexts. For example, Tracy and Dimock highlighted the three key meeting processes of (1) giving and getting information, (2) deliberating/problem solving/decision making, and (3) coordination and planning of future action.[28(p123),see also29] *Pedagogy* is not commonly included as a key meeting process within meeting science; however, training is often described as a *goal* of meetings,[30] and pedagogy is clearly relevant in rounding research set in teaching hospitals.[31]

RQ2: How is Cross-Team Coordination Acknowledged and Addressed During Cancer Care MTS Meetings?

Our thematic framework of health care meeting processes overlaps with insights found in existing studies of health care meeting processes and the broader literature on meeting science. However, the notable contribution of this framework most clearly emerged in response to our second research question: the identification of how MTS considerations are (or could be) incorporated during these meetings. In the following, we discuss two ways in which cross-team coordination can be acknowledged and addressed during cancer care MTS meetings.

Considering cross-team coordination in discussion. Throughout all four of the higher-order themes, we observed examples of meeting attendees leveraging these processes to facilitate cross-team coordination (see Table 1, especially subprocesses labeled activities in other teams). First, regarding *informing*, we observed many instances where attendees offered information related to their understanding of

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TABLE 1. Framework of Cancer Care Multiteam System Meeting Processes With Examples

Subprocess	Examples
Process 1: Informing—volunteering information about the patient's current and historical condition along with updates about care plan activities	
Patient history and context Informing about patients' personal history and context including their medical, physical, social, psychological, and economic circumstances	*Nurse navigator provides context of a patient's personal history, including losing house and job during treatment.*[a] Nurse navigator clarifies that [the] patient is getting [a] nephrostomy tube put in and it is painful (**R01**)[b] The group (registered nurse, nurse manager, and case manager) starts sharing information about what they know the patient's home situation to be like; *Case manager says she believes the patient has a younger sister that can come help [the patient at home]* (**H01**)
Patient status Informing about recent events and activities that elucidate patients' current health, medical standing, and progression	*Resident reports there was one barrier to the patient leaving—voiding on her own. They offered her discharge, but it was refused. The catheter is out, and she has voided on her own.* The goal is for the registered nurse [who was not present during these rounds] to get the patient out at 10 (**R02**) *Registered nurse gives an update about the next patient. Case manager adds that for this patient, they attempted to advance to a regular diet. The case manager also ordered a TPN [for the patient]* (**H02**)
Upcoming events and goals Informing about the current care plan and associated goals, events, and activities as they align with the patient's current care status	Resident gives today's plan for the patient. She said she put patients on prophylactic Lovenox. *Goals are to pull foley out, get ambulating, tolerate PO. Resident says she "thinks the patient needs to meet a few more goals but is moving in the right direction"* (**R03**) Registered nurse said [this patient is] post op day 6. [Patient has a] clamped NG tube. No nausea but pain. Patient told her chemo is tomorrow. *She said Sunday/Monday [will be the day she goes] home. [The charge nurse writes on the board] the day of discharge* (**H03**)
Activities in other teams Informing about the activities, status, or sentiments pertaining to, or put forth by, individuals in other hospital teams	Resident gives update: *pulmonology came to see [the patient] to assess how they think she would do for surgery. [Pulmonary] said [the patient] was a moderate risk because of her chronic steroid use she would need stress dose steroids if [they were to] operate* (**R04**) [About discharge for a patient] Registered nurse says that [the patient's attending] says he wants her more stable [before discharge], which could take many days. *[According to the registered nurse] the attending says he is going to get her some liquids to sip on* (**H04**)
Process 2: Deliberating—Exchanges among meeting attendees that involve raising and addressing issues that clarify, contribute, correct, and confirm information related to patient situations, to reach and establish next steps in care plans	
Patient history and context Exchanges focused on details related to personal history and context including their medical, physical, social, psychological, and economic circumstances	*Attending asked how [the patient] ended up in the hospital since she has an oncologist. [The registered] Nurse said she was at Hillcrest, which is in the hospital network* (**R05**) *The registered nurse wonders whether [discharge] vouchers might be needed for [a particular] patient. Case manager said, no, she's got Medicaid Blue choice* (**H05**)
Patient status Exchanges focused on details related to recent updates, events, and activities concerning patients' current status, the care plan, and progression	There is a back and forth between two residents and the attending. Resident points out that she [patient] is having increasing bowl movements again. Second resident says she's eating more, as a positive. *Attending asked if she had Laxatives yesterday. Resident says yes. Attending asks if the pathology [report] was back yet. Resident said that it wasn't back yesterday but that he hadn't checked this morning* (**R06**) The case manager asks [whether the patient received] "any dressing changes?" Registered nurse: "don't think so" (**H06**)
Upcoming events and goals Exchanges focused on details related to upcoming procedures, activities, and care plan progression and associated goals	[Rounding group had approached registered nurse to discuss a patient]. Attending asks: "anything you need from us?" *Nurse asks if they are going to place a drain today. Attending says no* (**R07**) There was one patient the group was confused about [regarding the care plan]. *The nurse manager double checked with the registered nurse that he understood, [the nurse said] the patient was admitted for surgery that wouldn't take place until 2 weeks from now* (**H07**)

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TABLE 1. Framework of Cancer Care Multiteam System Meeting Processes With Examples (continued)

Subprocess	Examples
Searching for information within and across teams Exchanges focused on searching for, or planning to search for, information needed to resolve issues and establish a care plan and goals	There are questions about whether the patient had gotten some images back, which would show if a tube had been properly placed in the patient. *Resident gets on her phone. Resident is reading something from her phone and seems to say that it [an image] had been ordered and [the image showed that] the tube was in the right spot* (**R08**) [Registered nurse was not available to attend huddle]. Nurse manager asks case manager to [give updates about the] absent nurse's patient. *Case manager is going through based on notes submitted. Case manager: "let's see if I can figure something out." Case manager is trying to figure out what happened on Friday. She reads verbatim some goals of care regarding the family.* Nurse manager adds that he was in with this patient they are talking about and that the patient said they are going to have surgery, but when he asked what kind, the patient didn't know. *Case manager searches for information in the notes about surgery and finds something [related to the kind of surgery the patient is getting]* (**H08**)
Activities in other teams Exchanges focused on details related to actions, activities, or sentiments put forth by individuals in other hospital teams	[Resident describes the] plan: lovenox teaching, walking around, hospitalist helping with hypertension. Resident says [we] just need to manage hypertension. *Attending asks about hospitalists again. Resident said they are consulted, and they had changed her regiment. Resident had asked hospitalists to assist with changing something to help with her blood sugars at home* (**R09**) *Registered nurse gave updates about the next patient. The nurse manager asked if the [patient's Attending] had seen this patient yet and the nurse said yes* (**H09**)
Process 3: Strategizing—Discussions focused on establishing goals and plans of care specifying specific future actions and coordination strategies	
Establishing care plans Deciding on the course of action to progress patient care and/or achieve specific goals	While we were walking the attending asks the registered nurse about [patient] number 28. Attending mentions letting her go home, and the registered nurse notes the patient was having headaches and passing gas. *[While in with the patient] notes he is checking in on the patient and will let her go home [today]* (**R10**) The registered nurse gives an update: "[the patient's] observation was 2 hours ago; *she [the patient] will be discharged today"* (**H10**)
Establishing contingencies Deciding on future goals or courses of action that may be pursued depending on certain contingencies	The resident said another concern is that the patient's fluids are not where they should be. The resident suggested revisiting the patient after rounds to check fluids again. *Attending said patient's urine output is not good but not horrible. If urine output increases, the catheter may be able to come out later today and discharge tomorrow if the patient can "shuffle around"* (**R11**) *[Registered nurse reports that] we'll plan for discharge tomorrow if [the patient's] pain [is] controlled and blood pressure [gets more] stable* (**H11**)
Adjusting care plans Making adjustments to the care plan and related goals on the basis of shared information or discussions	*Resident reports that they had everything ready for her [patient] to get discharged yesterday but surgery said they had the wrong wound vac set up at her home. They wanted variflow for this patient.* Attending said he never heard of that before and asks for more info about this variflow. Resident describes what it is. Attending asks resident to look more into it. Resident starts to say that we need to let case management know. Hopefully, she can go home today. Nurse navigator read in a note from a wound nurse that you change this out 3 times a week. *Attending asks if she can go home with the other version in the meantime. Residents both say no and explain why. Nurse navigator was like "we didn't know that" in an 'oh well' sort of tone. Resident says it's all good and we'll figure it out—reassuring* (**R12**) [About a patient who is getting discharged into an assisted living facility] Case manager says to the registered nurse that she will arrange a COVID-19 test for the patient. *The social worker says "hold off on that" because that would generate more paperwork for them. Case manager says ok, and then says to the registered nurse to never mind that then* (**H12**)

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TABLE 1. Framework of Cancer Care Multiteam System Meeting Processes With Examples (continued)

Subprocess	Examples
Coordination with other teams Planning interactions and activities that involve working with individuals on other teams and groups to achieve goals	[Speaking about a patient who is getting a nephrostomy tube placed]. Attending asked nurse navigator about scheduling next, and any "hold ups" [to the patient's discharge plan] she anticipates this creating. *Nurse navigator develops a plan to work with case manager to address any hold ups due to [nephrostomy tube placement]* (**R13**) Registered nurse [had been told by the patient that the patient's] chemo is starting tomorrow. She said Sunday/Monday [the patient will] go home. The registered nurse doesn't know if they are stopping [the patient's] TPN or if it will be cyclic. *The Dietary clinician [AKA nutritionist] notes that she will call them [the group handling TPN]* (**H13**)
Process 4: Pedagogy—Exchanges that educate meeting attendees in health care, medicine, and the procedures involved in health care meetings	
Teaching Taking opportunities for teaching, learning, lecturing, and quizzing about medicine and health care	*Attending then brings up the patient's TPN, he wants the residents to think about nutrition and balancing it. Attending starts quizzing resident about how many calories someone needs for their body weight and hints that there is a formula. Another resident guessed after some silence, but the formula he starts to recite was for urine output. The first resident could also not remember. Attending asks if she wants a lifeline with some laughing, the response from the group is yes* (**R14**) *[With registered nurse present] resident begins giving report on patient. Attending gives the resident feedback on the way he should deliver reports, pointing out to the resident that 'I don't know what it is you don't know'. The Attendees talked about the way presentations should be done: "tumor, timeline, treatment, disease status." Attending helps the resident fix his presentation and tells him to "try again"* (**R15**)

Abbreviations: PO, food by mouth; TPN, total parenteral nutrition.

[a]Italicized text in the illustrative excerpts directly illustrate the process subthemes. The surrounding unitalicized text acts as additional context.

[b]For each subprocess, two examples are provided, one from rounds [denoted with R] and one from huddles [denoted with H]. An exception is made for *teaching* as this subprocess was not observed during huddles.

the activities occurring in other component teams. Second, during *deliberating* processes, we observed back-and-forth exchanges that were intended to create shared understanding about how different teams have, or should, coordinate to support patient care. Third, within the *strategizing* theme, we observed several meetings where attendees articulated care plans that specifically incorporated how and when component teams would coordinate their future actions. Finally, regarding *pedagogy*, we observed a small number of meetings where attendees leveraged the meeting as an opportunity to provide training that facilitated shared understanding across teams. This can be seen when RNs were present during rounds (**R15**).

Meeting composition (component team representation). In addition to acknowledging cross-team coordination during informing, deliberating, strategizing, and pedagogical processes, we also observed instances when *meeting*

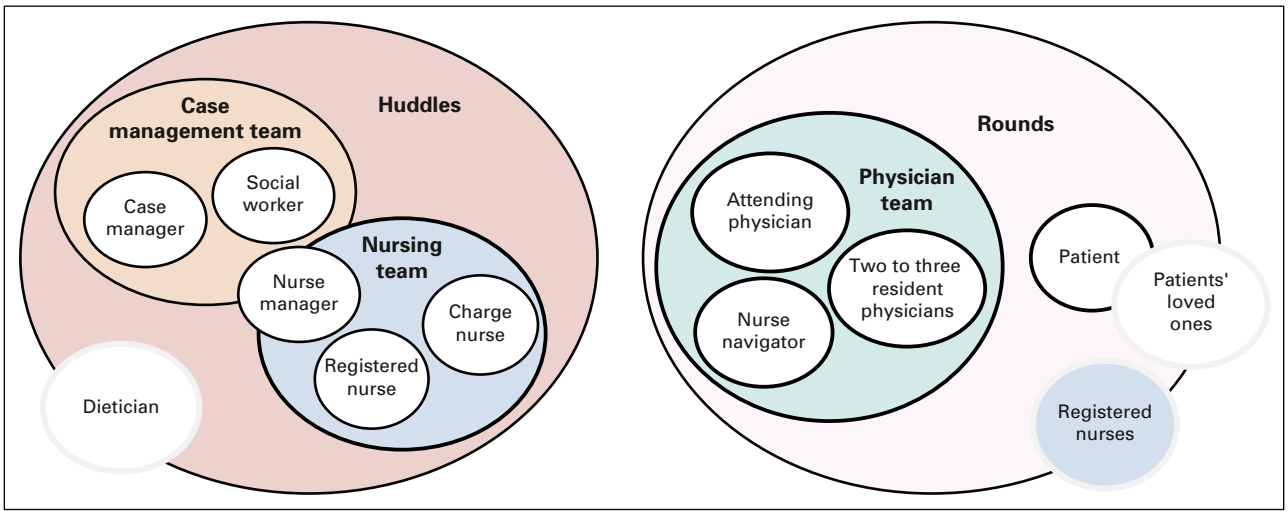

FIG 1. Typical composition of rounds and huddles through a multiteam system lens. A depiction of the teams, and individual health care professionals, typically involved in huddles and rounds. Health care providers circled by gray lines, as apposed to black lines, were occasional, inconsistent members of these meetings.

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composition (who is in attendance) facilitated or limited MTS coordination. As noted previously in RQ1, the meetings varied with regard to which component teams were represented. Rounds were typically only attended by physician team members, which granted them an opportunity to coordinate *within* their team (rather than opportunities to coordinate with members of other teams). On the other hand, huddles were opportunities for cross-team coordination between the nursing and case management team members. None of the meetings we observed included representation from *all* component teams in the system, which might have limited cross-team coordination.

Notably, in many of the meetings, attendees engaged in informing, deliberation, or strategizing processes that referenced members of teams that were *not* in attendance (eg, **H04** and **R13**). Indeed, some meeting attendees leveraged their attendance across *multiple* meetings to establish information channels between teams. Consider the following example (not included in Table 1) showcasing such an event, where an RN relayed information from rounds to huddles:

> [During rounds] the AP asks the RN about [patient] number 28. AP mentions letting her go home & the nurse notes the patient was having headaches & passing gas.
> [Later during huddles] the RN gives an update: "[the patient's] observation was 2 hours ago. She will be discharged today."

The previous example can be juxtaposed with the following example where the RN was absent during rounds and did not overhear the AP informing the patient that she may be discharged today if able to tolerate food. In this second case, information generated during rounds was not communicated to attendees during the huddle, and the RN was not aware of recent developments with the patient.

> [AP tells a patient] we'll let you get home this afternoon if all goes well' … "Let's see how you do with breakfast and lunch off your pain meds."
> [Later during huddles] The RN reports that the patient is now without ketamine. There was some question about whether she could have lunch or not—it was not clear whether this was resolved.

DISCUSSION

Our qualitative investigation of cross-team coordination in inpatient cancer care MTSs revealed that MTS meetings such as rounds and huddles have significant potential as cross-team coordination tools. These meetings provide formal, routinized opportunities for key cross-team coordination processes, each of which included subprocesses that highlight important opportunities for supporting cross-team activities. However, although there are *opportunities* for facilitating cross-team coordination processes during meetings, our interviews and observations revealed several *challenges* that can make successful facilitation of cross-team activities in meetings more difficult. Fortunately, existing meeting science,[1,29,32,33] MTS,[15,34-37] interdisciplinary rounding,[3-5,22,24] and literature studies provide a rich source for evidence-based recommendations. Table 2 summarizes the challenges and opportunities we identified, organized by process and paired with evidence-based considerations for clinicians wishing to make their huddles, rounds, and other MTS meetings more effective for cross-team coordination.

First, our study reveals the opportunity of rounds and huddles to serve as well-structured, routinized meetings that ideally involve representation from each of the core component teams. Rounds and huddles were identified as one of the very few times that members are together in a synchronous, face-to-face setting. In alignment with the meeting science and MTS literature studies, it is important that meeting attendees are provided with opportunities to share information, engage in deliberations, and contribute to strategies, especially if serving as the sole representative from their team.[30,32,34,35,41]

However, there is also the challenge of composition. It is critical to consider what aspects of MTS meetings may not be relevant or appropriate for all members, to avoid instituting meetings that might be inefficient, irrelevant, or dissatisfying.[1,29,42,60] For instance, pedagogy is an important process for physician teams during rounds, but less so for nursing or case management teams. MTS leaders should consider structuring meetings to consider MTS-relevant coordination needs, emphasizing how meeting involvement can support activities within and across teams. For rounds, this might occur by incorporating other teams' representatives during prebrief activities that prepare teams to coordinate in upcoming patient care (eg, strategizing), while using debrief activities for pedagogy, reviewing past events to support learning and coordination improvements (eg, informing).

Another opportunity comes from our finding of RNs and NNs taking on the role of an informal liaison for component teams who were not able to be present during MTS meetings. In the MTS literature, this is referred to as a *boundary-spanning role,* which involves a member of one team navigating relationships with another team in terms of sharing information, planning, and strategizing for the greater benefit of the MTS as a whole.[15,61] Although evidence-based practices suggest that MTS leaders align component teams' schedules and work activities, so they can attend one another's meetings,[34,44] realistically, this may be difficult in the complex scheduling of health care. An alternative, evidence-based practice is to identify meeting attendees who can serve as boundary spanners to communicate pertinent information across teams before (or after) MTS meetings.[44]

Our research applied a qualitative, single-site case study in an inhospital setting, limiting the generalizability of findings. However, given the growing prevalence of health care MTSs and the need to understand their cross-team coordination,[18,19]

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TABLE 2. Evidence-Based Practical Considerations for Promoting Cross-Team Coordination During Inhospital Cancer Care MTS Meetings

Process	Key Cross-Team Coordination Challenges and Opportunities Identified During MTS Observations and Interviews	Associated Evidence-Based Practical Considerations
Informing	***Opportunity***: rounds and huddles were identified as one of the very few times that members from component teams have the chance to share information in a synchronous, face-to-face setting	Ensure that every meeting attendee is provided with the opportunity to share information, especially if serving as the sole representative from their team.[34,35,38-40]
	Opportunity: bringing together representatives from multiple component teams can provide opportunities to offer information in relation to other teams' activities, especially information that they would not have realized was relevant without hearing about those activities	When deciding who should attend the meeting, consider which component team members/roles may be most relevant or critical for sharing information.[1,21,30] Structure meetings to include time explicitly for cross-team information sharing, ideally when most or all component teams are represented.[21,29,30]
	Challenge: None of the meetings observed included a representative from every component team, limiting the opportunity for comprehensive cross-team information sharing	Attempt to schedule rounds and huddles at times when the greatest number of component teams can be represented.[30,41,42] If it is not possible for a team to be represented, establish an alternative option (eg, a nurse navigator who can serve as a boundary spanner who can relay information to and from the component team).[15,35,43]
Deliberating	***Opportunity***: MTS meetings such as rounds and huddles allow for component team members to follow up and ask questions in a synchronous, face-to-face setting	Allocate time so that every meeting attendee has an opportunity to ask questions and clarify information, especially new team members.[33,42,44-47] Include explicit opportunities for clarifying activities that are happening within different teams.[15,36,37,48-51]
	Challenge: If a component team is not represented in rounds or huddles, they are unable to provide responses to questions that come up during deliberations, which can affect later strategizing or may delay decision making	Use supporting resources in real time (eg, examining medical records) to help with addressing questions during deliberations, particularly if not all component teams are represented in the meeting.[30,34,40,49,52,53]
	Challenge: There may be some discomfort in speaking up or questioning the information provided during huddles and rounds, especially in terms of cross-team deliberating	As a meeting leader, create a psychologically safe environment by admitting when something is not clear, asking questions, and inviting individuals to speak up.[29,31,52,54-56]
Strategizing	***Challenge***: There are some strategy discussions during rounds and huddles that are only relevant for a subset of component team members	When establishing or revising plans for specific patients, consider whether meetings should involve representatives from the entire MTS or only certain component teams.[1,16,29,30,42,45,49,51]
	Opportunity: Huddles and rounds provide opportunities to develop strategies that take into account multiple teams' activities, especially in terms of the timing of different events	As meeting leader, ensure that strategizing discussions allow for cross-team activity planning, especially the specific timing of activities.[21,49,50]
	Challenge: When a component team is not represented at rounds or huddles, there may be a delay in them receiving updates to the strategies and plans discussed in the meeting	If a component team is not represented during a strategizing discussion, ensure that someone who is attending the meeting is assigned the action of following up with that component team to relay the plan and address any additional questions or contingencies.[15,21]
Pedagogy	***Challenge***: Rounds are intended to be a learning opportunity, so attendance is most advantageous to RPs and learners from other component teams (eg, newly hired nurses) and less so for nonlearners	During rounds, consider conducting pedagogically focused activities when learners are available from more than one component team (eg, RPs and newly hired RNs).[29,30,57,58] Plan meeting schedules and agendas to prioritize the time of team members who are necessary for other meeting processes (eg, informing, deliberating, and strategizing) but may be less likely to benefit from pedagogically focused activities.[1,21,29,30,45,59]

Abbreviations: MTS, multiteam system; RN, registered nurse; RP, resident physician.

we encourage further empirical research that can aid in systematically uncovering the impact of MTS meetings on cross-team coordination. This would extend previous research suggesting that cancer care meetings (particularly interdisciplinary ones) fortify efficiency by bolstering implicit coordination.[23,31]

In addition, our focus on understanding processes meant that we are limited in addressing the impact of MTS meetings on outcomes. Future research should consider the impact of cancer care MTS meeting design on health care efficiency and performance metrics[30,32,41,60] such as length of stay,[18] errors/near misses,[19] and communication breakdowns.[10,32] We also encourage future research that considers how increasing component team representation in meetings to facilitate MTS performance must be balanced with the need to prevent work overload and employee burnout.[1,45,62]

A final limitation of our research is the focus only on inpatient hospital settings. MTS structures undoubtedly exist in other areas of cancer care delivery, including outpatient[18,63] and

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the more complex outpatient-inpatient MTS.[64] Future research is warranted to better understand MTS meetings and their cross-team coordination processes in these similar yet different MTS settings.

In conclusion, although rounds and huddles are certainly not new,[3-6,22,23,31] by leveraging an MTS lens, we have uncovered their value to facilitating cross-team coordination during inpatient cancer care delivery. Our qualitative approach has uncovered a richer understanding and initial framework that we hope will be further explored and expanded with empirical research endeavors in additional settings. Finally, we hope that our resulting framework and associated practical considerations can offer meaningful guidance to clinicians who would like to better use their rounds and huddles as tools to support cross-team coordination.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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Manuscript writing: All authors
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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Making Every Meeting Count: A Qualitative Investigation of Multiteam Meeting Events and Their Role in Supporting Coordinated Cancer Care Delivery

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APPENDIX 1. SEMISTRUCTURED INTERVIEW GUIDE AND STARTER QUESTIONS

Introduction and Background

Hello, my name is ____. Thank you for joining me today for this interview.

As you might know, this research is focused on improving collaboration between the different teams who share patients on [unit/department]. We are especially interested in how collaboration between teams on [unit/department] might be facilitated by interdisciplinary patient rounding (ie, rounds that include physicians, residents, and bedside nurses in discussions of a patient's care).

We would ultimately like to develop theories and recommendations that can be used to maximize the effectiveness of interdisciplinary rounds to help prevent collaboration breakdowns between teams and ultimately lead to better patient outcomes in terms of efficient discharge, lower length of stay, fewer preventable infections, and efficient communication and collaboration between providers.

Multiteam System Questions

In team science research, we think of interdependent sets of two or more component teams that need to work together fairly closely to accomplish a shared goal, as a *multiteam system (MTS)*.

BASED ON THIS DEFINITION, WOULD YOU CONSIDER THE CARE OF PATIENTS ON [unit/department] TO REQUIRE MULTITEAM SYSTEMS?

- If so, who else is part of this MTS generally? (eg, nurses, surgeons, residents, pharmacy, techs, unit secretary, and nurse navigators)

COULD YOU TELL ME WHAT TEAM (OR TEAMS) YOU BELONG TO IN REGARD TO PATIENT CARE ON [unit/department]?

- What other teams do you rely on?
- What are the primary objectives of the teams you belong to? What are the ultimate outcomes/goals you want to achieve for a patient?
- What about the objectives of the entire MTS? What goals are shared across teams?
- Which people on your team tend to be involved with communicating and coordinating with other teams?

Interdisciplinary Rounding Questions

The way we are defining interdisciplinary rounding is a round that includes more than one discipline, such as nurses and physicians, discussing their patient together and possibly visiting the patient together at the bedside.

BASED ON THIS DEFINITION, HOW FAMILIAR YOU ARE WITH INTERDISCIPLINARY ROUNDING ON [unit/department]? DO YOU SEE IT TAKING PLACE VERY OFTEN, OR ARE YOU INVOLVED WITH IT?

- When does it occur?
- Who tends to be involved? Who tends to conduct it?

WHAT PURPOSES DO YOU SEE DAILY INTERDISCIPLINARY ROUNDS AS SERVING?

- Does information about what has been discussed during IDR make its way back to you? How?

YOU LIKELY MAKE A LOT OF TYPES OF ROUNDS TO PROVIDE PATIENT CARE. COULD YOU DESCRIBE WHAT THE PURPOSES OF THE ROUNDS ARE THAT YOU DO?

- Do you have the opportunity to conduct interdisciplinary rounds with physicians at the bedside? (nurses)

WHAT IS YOUR ROLE WITH REGARD TO PATIENT INTERDISCIPLINARY ROUNDING?

- What are you usually doing and hoping to achieve during rounds?
- Besides the patient and their families, do you find yourself interacting with people outside of who you consider to be on your team?
- To what extent are the rounds you make interdisciplinary?
- Does it involve involvement or input from other teams during the rounds?
- What about before or after?
- Do you prepare for rounds by consulting members of other teams? Or do you transfer that information to other teams after rounds?

FROM YOUR PERSPECTIVE AS A _____, WHAT ARE THE BENEFITS OF NURSE PARTICIPATION IN INTERDISCIPLINARY ROUNDING?

- How can you tell that there are benefits? What lets you know this?
- Are there any downsides?
- Why, and for which teams, are these things beneficial?

FROM YOUR PERSPECTIVE AS A _____, WHAT NEEDS TO HAPPEN IN ORDER FOR NURSES TO PARTICIPATE IN IDR MORE CONSISTENTLY?

- What are the logistic barriers?
- What are the behavioral barriers? From nurses or attendings, residents, medical students, etc?
- What changes could be made to make interdisciplinary rounding more frequent?

IF NURSES DO NOT PARTICIPATE IN IDR, HOW DO TEAMS CONNECT WITH EACH OTHER OVER INFORMATION THAT WAS DISCUSSED DURING THE ROUNDS? HOW IS THAT GAP CLOSED?

- Who is the person/people on your team that usually communicates with other teams about this information?
- Through what medium?

IF YOU DON'T TAKE PART IN INTERDISCIPLINARY ROUNDING, WHAT OTHER SORTS OF ROUNDING DO YOU DO?

- What are your role and goal there?
- Does it involve involvement or input from other teams during the rounds?
- What about before or after? Do you prepare for rounds by consulting members of other teams? Or do you transfer that information to other teams after rounds?

APPENDIX 2. EXPANDED DESCRIPTION OF THE QUALITATIVE ANALYSIS SCHEME FOR OBSERVATIONAL DATA

Initial coding: First, three coders individually conducted initial coding, which entailed going through observational transcripts line-by-line to apply initial codes. In this stage, gerunds, which are action words ending in *-ing* (ie, *discussing* treatment plan and *teaching*), were used as coding labels. Use of gerunds for initial coding is recommended as

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best practice in constructivist approaches to qualitative research because it limits subjective data interpretation and highlights observable processes within the data.[25(p221-227)]

Focused coding: Second, the lead researcher engaged in focused coding[25(p247-282)] to identify higher-order themes from the initial codes. This involved comparing and contrasting the initial codes to identify patterns and themes that were more common and significant. Comparisons were made across codes generated for different meetings (i.e., rounds and huddles) and between different coders analyzing the same events. Higher-order codes were then translated to themes and integrated into a framework that described key cancer care meeting processes and subprocesses. This stage of the coding scheme was facilitated within MAXQDA qualitative analysis software.

Recoding/consensus coding: In the spirit of abductive reasoning, where data are often revisited and reinterpreted in new light,[25(p341-347)] we reapplied codes within the thematic framework to the observational data. This constituted a third stage of analysis intended to assess whether all aspects of the framework were applicable to both rounds and huddles and to ensure cross-coder agreement on the final framework.

Round and huddle transcripts were revisited by the lead researcher, who identified 50 sections from field observations and coded these using the processes and subprocesses within the initial thematic framework. The 50 sections accounted for 12% of transcripts for rounds and 11% for huddles. Of these 50 sections, the lead author selected examples from rounds and huddles for each of the thematic framework's process-subprocess combinations, aside from the *pedagogy* process where two examples were taken from rounds due to there being no observations of the teaching subprocess within huddles. The resulting 28 sections were recoded by the two other analysts who had been involved during the initial coding phase. To assess the agreement between the three coders in applying processes and subprocesses from the thematic framework to 28 examples of cancer care meetings, Fleiss kappa was computed; the statistic suggested good intercoder agreement (ie, kappa $= 0.731$, $P < .0001$).

Coders held joint meetings to discuss any of the 28 examples that received conflicting codes. Coders ultimately reached 100% agreement in their application of the process framework to the 28 examples through joint discussions, which resulted in minor edits to the framework and its definitions.

Coder training: Three coders were involved in the qualitative analysis, and these individuals also served as observers during data collection. Coders were given training on the constructivist approach to coding and given the opportunity to practice line-by-line coding before actual data analysis. Coders were encouraged to approach the data initially using *sensitizing concepts* reflective of multiteam system and team theories. Charmaz describes sensitizing concepts as "give[ing] researchers initial but tentative ideas to pursue and questions to raise about their topics … [they] provide a place to start inquiry, not to end it."[25(p84)] In this research, concepts include team(s), multiteam system goals, and between-team and within-team coordination.

Coders conducted a practice round of line-by-line coding on a sample of observational data that was not included in the final analysis. Coders were trained to use gerunds,[25(p221-227)] action words ending in -ing (ie, *discussing* the treatment plan and *teaching*). Gerunds provide a fruitful way to limit subjective data interpretation by highlighted observable processes and actions that were ongoing during meetings. After individually completing the practice sample, coders discussed the process and outcomes of the practice to identify similarities and differences in codes. In so doing, the coders calibrated and aligned their approaches to coding.

Coder meetings: During analysis, coders held consensus meetings regularly through all stages to resolve intercoder discrepancies and make amendments to the thematic framework where needed. In the spirit of the constructivist approach taken during this qualitative analysis, it was expected that differing perspectives, life experiences, and professional backgrounds would generate different and contrasting interpretations of the data. These differences were scrutinized during meetings to generate new insights and adjust the coding approach and the emerging thematic framework. Coders had meetings and discussions about the usefulness and fit of the framework to the data, until unanimous consensus was reached.

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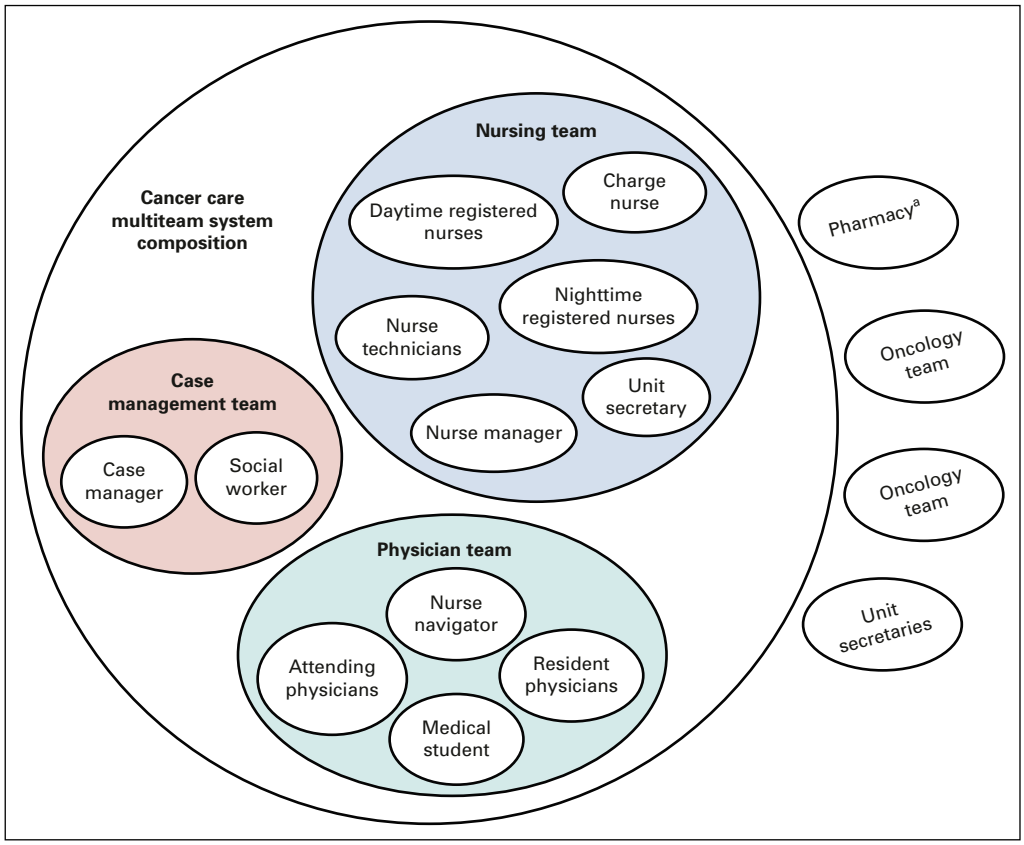

FIG A1. MTS structure within the hospital research context. A depiction of the MTS operating within the inhospital gynecologic oncology context under investigation. The MTS includes three core component teams, the case management, nursing, and physician teams. The patient and patient's loved ones serve as another team, not included in this research. Also identified are the individual health care professionals and hospital staff included in each component team. [a]In addition to the pharmacy group, other consulting teams, such as pulmonary, radiology, cardiology, and urology, were described as auxiliary teams in the MTS. MTS, multiteam system.

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